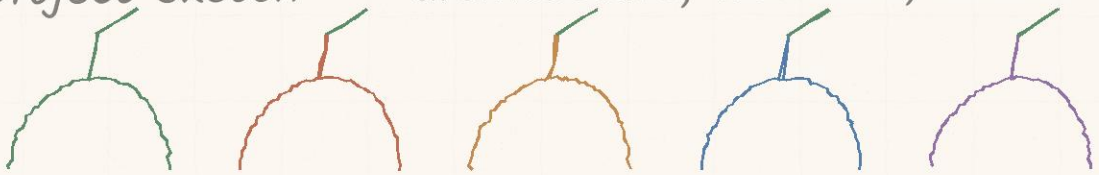
S.L.A Fruit Shop

project sketch - architecture, frontend, backend, flows



A 3D fruit storefront with e-commerce,
social login, checkout and an admin dashboard

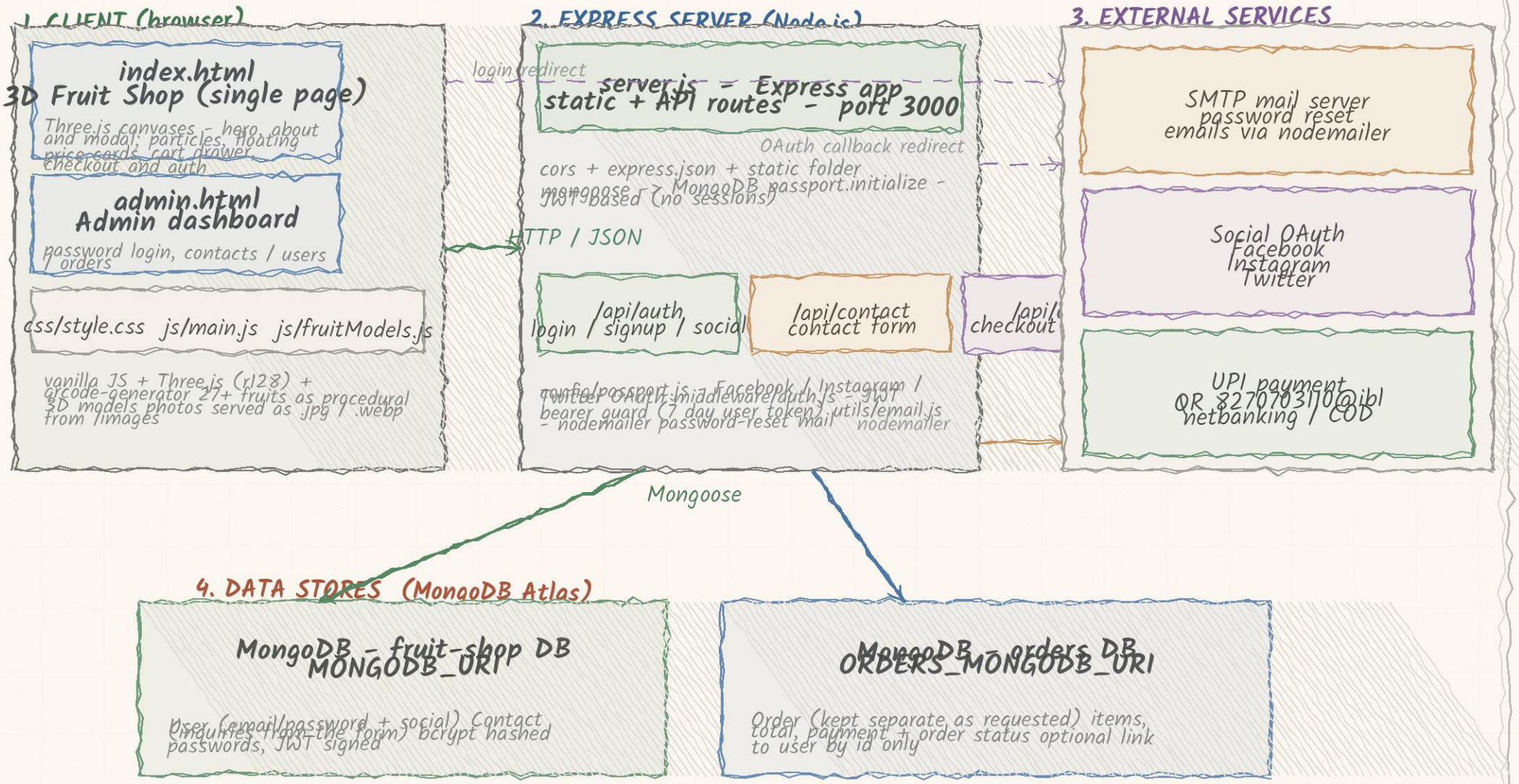
ECR Road, Kattumavadi - Tamil Nadu 614630

call 9715840263 - delivery within 3km

Node.js + Express + MongoDB + Mongoose + JWT + Passport + bcrypt
Three.js 3D + vanilla JS + nodemailer + qrcode-generator

S.L.A Fruit Shop — System Architecture

full-stack sketch: 3D fruit storefront + e-commerce + admin panel



Frontend Map — what the user sees

index.html storefront sections + floating widgets + admin panel

index.html — the 3D storefront (single page)

1 Loader + particles + marquee bar
animated loading, canvas background

2 Navbar + mega menus
sign-in, cart, call, mobile menu

3 Hero — Three.js 3D canvas
floating price cards, scroll cue

4 Trust bar
fresh / quality / free delivery

5 Categories
Indian, Imported, Seasonal, Wholesale

6 Services
wholesale, retail, delivery, imported

7 Indian fruits (18)
season + region filter tabs

8 Imported fruits (14)
continent filter tabs

9 Health benefits
Ayurvedic immunity / digestion / heart / skin

10 CTA banner
wholesale bulk orders, 2 phone numbers

11 Seasonal calendar
12-month fruit peak chart

12 About + stats
3D canvas, animated counters

13 Testimonials
customer reviews

14 Contact form
POST /api/contact

15 Footer + photo credits
address, hours, map link

Floating widgets & modals

- **Cart drawer** — add / remove / qty, total, call-to-order
- **Checkout modal** — UPI QR + netbanking + COD
- **Fruit modal** — 3D fruit model, price, nutrition bars
- **Auth modal** — sign-in, sign-up, forgot + reset password
- **Social logins** — Facebook, Instagram, Twitter buttons
- **User menu** — welcome name + sign out

Client assets

js/main.js js/fruitModels.js js/admin.js css/style.css
images/ .jpg images/ .webp Three.js r128 + qrcode-generator from CDN

Admin panel (admin.html)

/admin
admin password -> JWT 24h

Contact inquiries from /api/contact

Registered users from User model

Orders from orders DB

tab counts update on login

Backend – API routes + data models

Express endpoints, JWT guards and MongoDB schemas

Server core

server.js
static + routes
mongoose + passport

middleware/auth.js
JWT bearer guard
optionalAuth requireAuth

config/passport.js
Facebook / Instagram / Twitter
enabled only if keys in .env

utils/email.js
nodemailer reset mail

/api/auth

- ~ POST /signup
- ~ POST /signin
- ~ GET /me
- ~ POST /forgot-password
- ~ POST /reset-password
- ~ GET /facebook + /callback
- ~ GET /instagram + /callback
- ~ GET /twitter + /callback

/api/contact

- ~ POST / (save inquiry)
- ~ GET / (list all)

/api/orders

- ~ POST / (checkout, guests ok)
- ~ GET /mine (signed-in history)

/api/admin

- ~ POST /login (ADMIN_PASSWORD)
- ~ GET /contacts
- ~ GET /users
- ~ GET /orders
- ~ all guarded by adminAuth (JWT admin)

MongoDB models (Mongoose)

User

fruit-shop DB

- ~ name, email (unique), phone
- ~ password (bcrypt, local only)
- ~ provider: local / facebook / twitter / instagram
- ~ providerId, avatar
- ~ resetPasswordToken + Expires

Contact

fruit-shop DB

- ~ name, phone
- ~ service: retail / wholesale / delivery / other
- ~ address, message
- ~ createdAt

Order

orders DB (separate)

- ~ userId (soft ref, different DB)
- ~ customerName, email, phone, address
- ~ items: [{ name, price, unit, qty }]
- ~ totalAmount, paymentMethod: upi/netbanking/cod
- ~ paymentDetails { upiId | bankName }
- ~ paymentStatus: pending / paid / failed
- ~ orderStatus: placed ... cancelled

JWT: user 7 days | admin 24h | bcrypt salt 10 | orders kept in a separate DB

User Journeys — how people actually use the site

five end-to-end flows, from browsing to admin review

Journey 1 — shop & order



Journey 2 — account & social login



Journey 3 — password reset



Journey 4 — contact the shop



Journey 5 — admin review

